

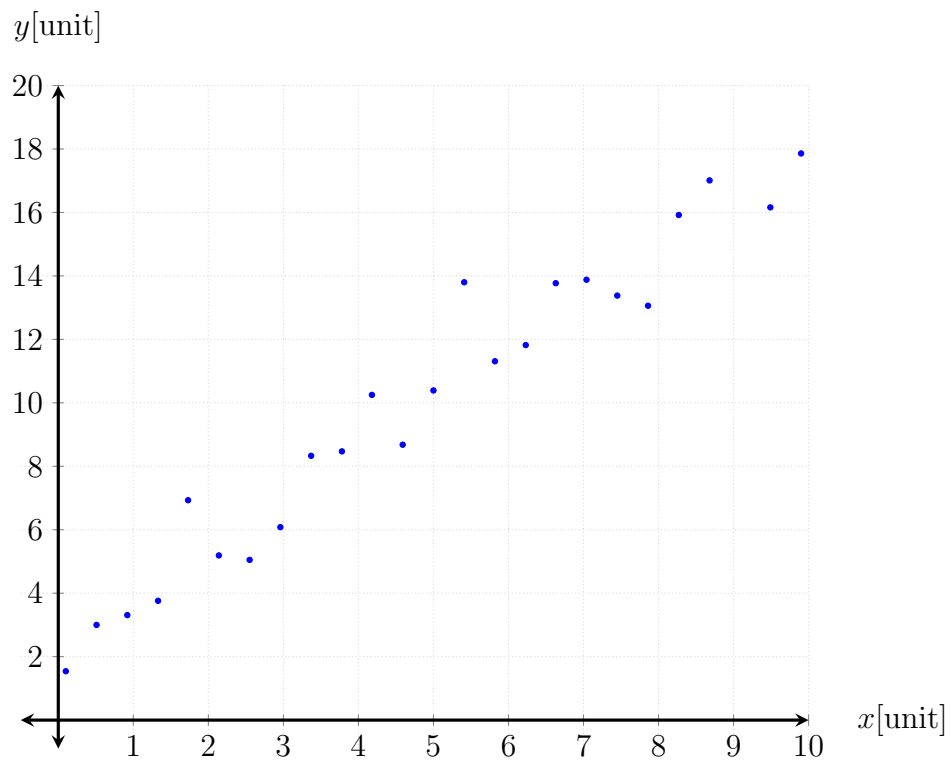
# Least Squares Method - A Quick Overview

Peleg Sapir

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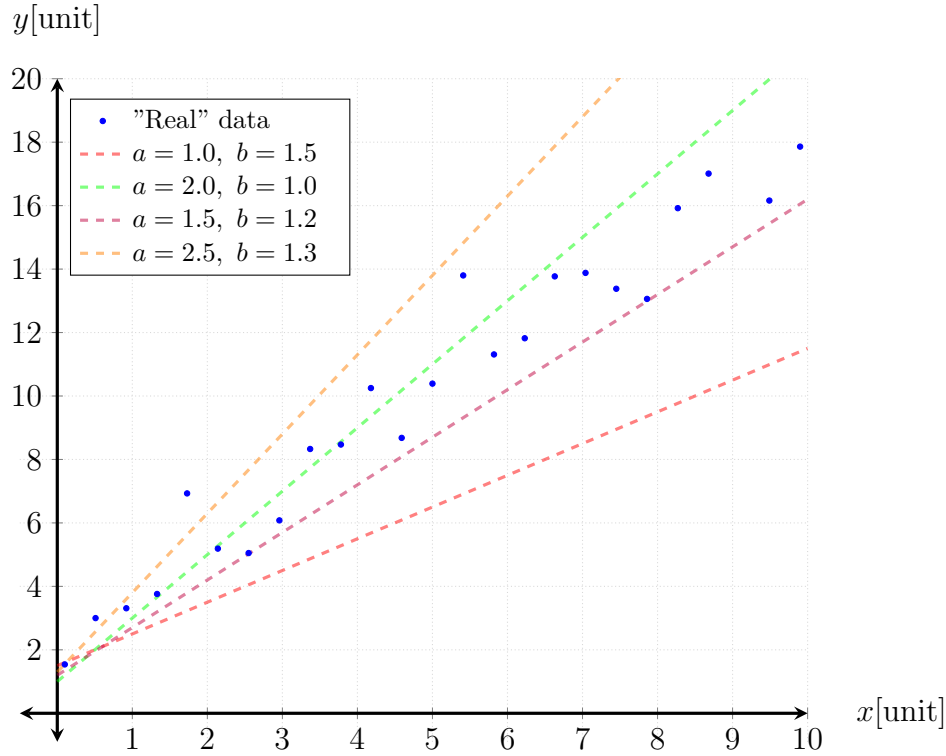
## 1 Stating the Problem

Consider the following set of points gathered in some experiment:



The distribution of the points seems linear, and we would like to know the approximate relation between the dependant variable  $y$  and the independant variable  $x$ .

How can we determine the *best approximate fit* for all the given points? Here are some examples of possible line fits, with their corresponding equations:



## 2 Finding the Best Fit

### 2.1 What we're minimizing?

The line of best fit has the formula  $y = mx + b$ , and we want to find the coefficients  $m$  and  $b$ . It's impossible to have a single line that would fit all points<sup>1</sup>. Instead, we will try to find the line that minimizes the total  $y$ -distances between the data points and the line itself.

(HERE BE DRAWING)

In equation form, each point has the form

$$p_i = (x_i, y_i), \quad (1)$$

where there are  $n$  points (i.e.  $i$  goes from 1 to  $n$ ).

The vertical distance between each point and the line  $y = mx + b$  is

$$r_i = y_i - (mx_i + b), \quad (2)$$

and the expression we want to minimize is the sum of the squares of these distances:

$$S = \sum_{i=1}^n r_i^2. \quad (3)$$

(we will see later why this expression is a good candidate for minimization)

### 2.2 Surprise appearance of linear algebra

If the line fit *every point*, we could write each of the points as

$$mx_i + b = y_i. \quad (4)$$

This is a set of  $n$  linear equations, which we can write as the following matrix-vector equation:

$$A\vec{v} = \vec{y}, \quad (5)$$

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<sup>1</sup>assuming it has zero width, of course.

or in explicit form:

$$\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} \quad (6)$$

However, this system has no solution (otherwise we could find the *perfect* fit), and so we will go for the next best thing: finding the projection of  $\vec{v}$  onto the subspace spanned by  $A$ , which we would call  $\tilde{v}$ . This  $\tilde{v}$  would be the minimization of the distance between the  $\vec{y}$  and  $A\tilde{x}$ , i.e.  $\min \{\|\vec{y} - A\tilde{x}\|\}$ , which is the same as finding  $\min \left\{ \|\vec{y} - A\tilde{x}\|^2 \right\}$  (since the square function is an injection over  $x \geq 0$ ).

The square norm  $\|\vec{y} - A\tilde{x}\|^2$  has the explicit form

$$\|\vec{y} - A\tilde{x}\|^2 = (y_1 - u_1)^2 + (y_2 - u_2)^2 + \cdots + (y_n - u_n)^2 = S. \quad (7)$$

TW: the projection part.